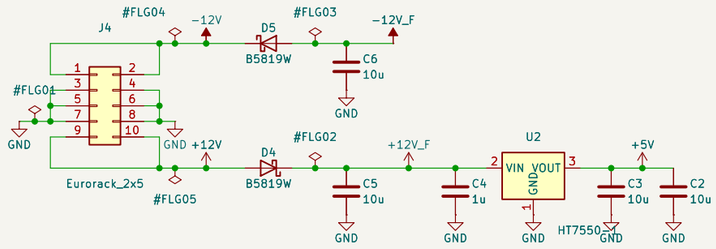
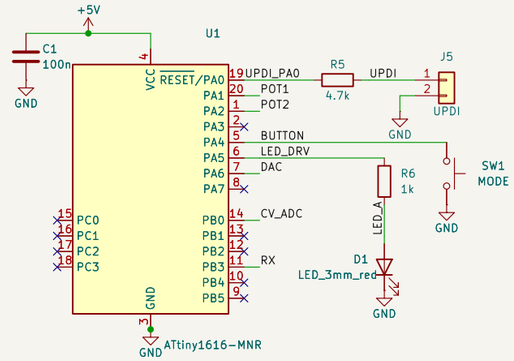
BIG MODULAR 8BIT — S4 schematic (Bread Modular 8bit for Eurorack, 4HP). Circuit intent: docs/research.md

POWER: +12V -> B5819W -> +12V_F (U3 V+, HT7550-1 -> +5V); -12V -> B5819W -> -12V_F (U3 V-).

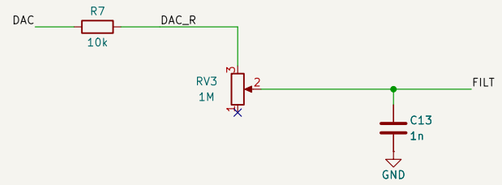


PWR_FLAG: #FLG01/#FLG04/#FLG05 mark the Eurorack GND/-12V/+12V entry at J4; #FLG02/#FLG03 mark the protected rails, fed only through D4/D5, which ERC cannot see through.

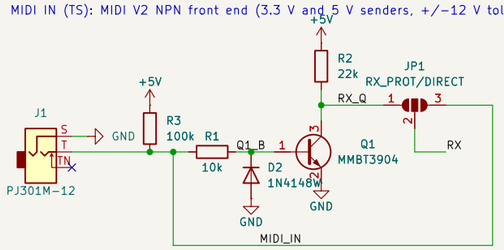
MCU: ATtiny1616 at 5 V: Bread Modular 8bit pin map (PA1/PA2 knobs, PA4 button, PA5 LED, PA6 DAC, PB3 RX). New: PB0 CV in, PB5 spare.
UPDI J5: pin 1 UPDI, pin 2 GND (2.54 mm). Power the module from Eurorack while programming.



KNOB 3 = PASSIVE LOW-PASS (as Bread Modular): DAC -> R7 10k -> RV3 1M rheostat -> FILT, C13 1n.
CW = brighter: 158 Hz .. 15.9 kHz. RV3 pin 1 open. U3A + input is high-Z, so the filter is unloaded.

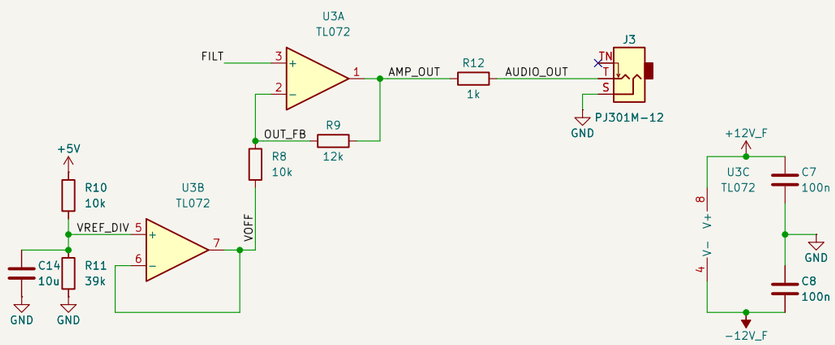


MIDI IN (TS): MIDI V2 NPN front end (3.3 V and 5 V senders, +/-12 V tolerant), no opto

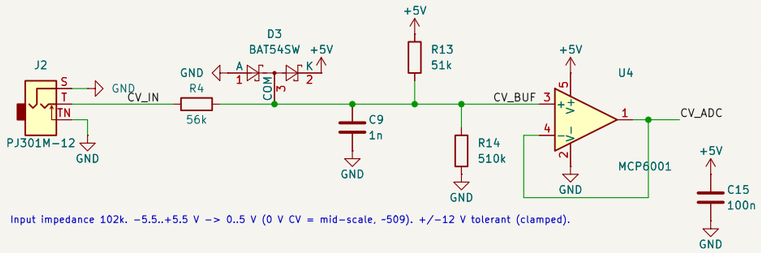


JP1 RX SOURCE (solder jumper on the back, shipped OPEN: bridge one pair by hand):
Solder 1-2 = PROTECTED (Q1 drives PB3; firmware PORT_INVEN on PB3).
Solder 2-3 = DIRECT (MIDI_IN to PB3, unprotected; stock BM 8bit firmware).
Exactly one bridge: never both, and open = PB3 floats.

OUTPUT: U3A gain 2.2 with -1.2 x VOFF (3.98 V): DAC 0..4.34 V -> -4.8..+4.8 V,
1k series to the jack. DC coupled (LFO firmware works).

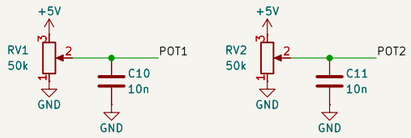


CV IN: R4/R13/R14 level shift, D3 clamp, C9 -> U4 MCP6001 follower (+5V) -> PB0 (AIN11)



Input impedance 102k. -5.5..+5.5 V -> 0..5 V (0 V CV = mid-scale, -509). +/-12 V tolerant (clamped).

KNOBS 1/2: 50k linear dividers +5V..GND into PA1/PA2 (BM CV1/CV2).



Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S4 — Bread Modular 8bit for Eurorack: protected MIDI in, bipolar buffered CV in, knob 3 pa

Big Modular

Sheet: /
File: 8bit.kicad_sch

Title: Big Modular — 8bit

Size: A3	Date:	**Rev: S4**
KiCad E.D.A. 10.0.6		Id: 1/1